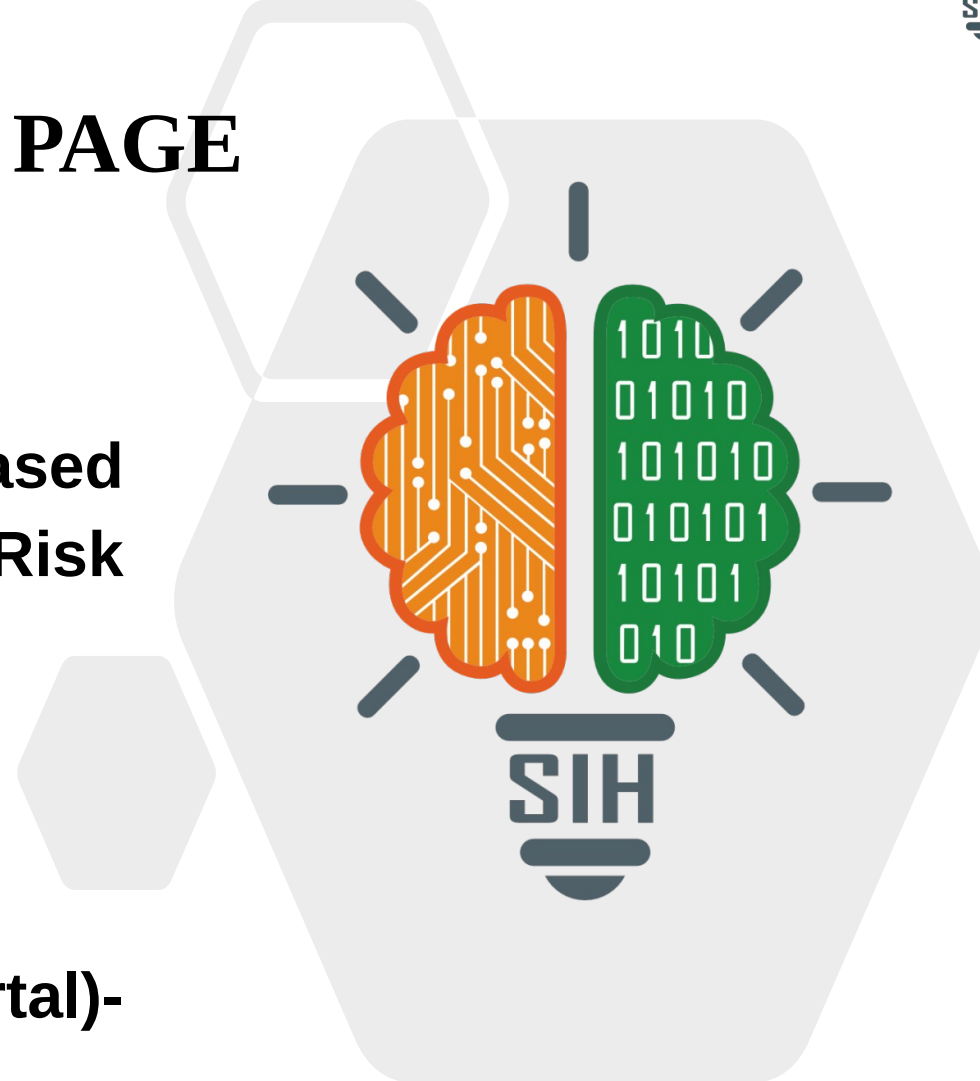
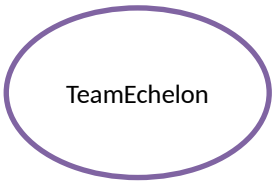




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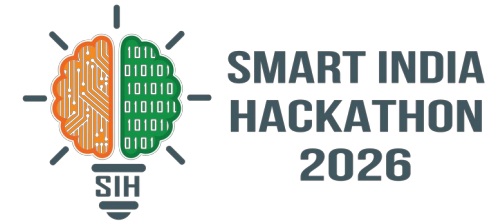
- Problem Statement ID – 26001
- Problem Statement Title- AI-Based Landslide Early Warning & Risk Monitoring System (NER)
- Theme- Disaster Management
- PS Category- Software
- Team ID- [Insert Team ID]
- Team Name (Registered on portal)- TeamEchelon





NER-Vision: AI Landslide Guardian

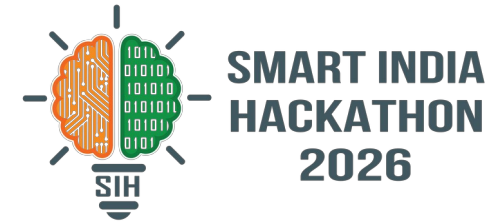
IDEA TITLE



❖ Proposed Solution (Describe your Idea/Solution/Prototype)

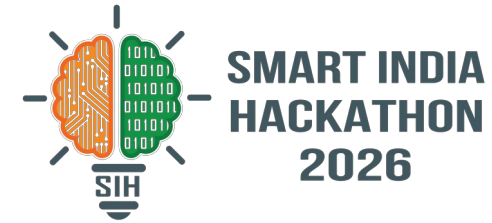
- AI/ML platform fusing rainfall, soil, satellite & terrain data to predict landslide risk in real time
- Shifts NER from reactive monitoring to proactive AI-driven early warnings, before a slope fails
- Explainable AI scoring, offline rule-based fallback, and risk-aware evacuation routing

TECHNICAL APPROACH



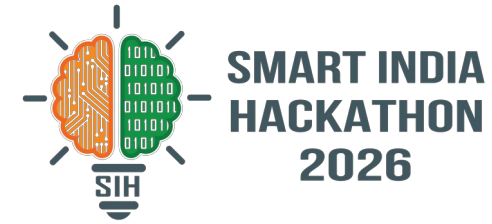
- Python/Flask backend, scikit-learn RandomForest risk model, NetworkX routing, Leaflet.js GIS dashboard; production adds IMD, Sentinel/Bhuvan, IoT sensors, Twilio
- Data ingestion → ML risk engine + offline rule fallback → GIS dashboard & alerts → crowdsourced reporting → evacuation routing. Prototype: 16 NER villages, 80% backtest accuracy

FEASIBILITY AND VIABILITY



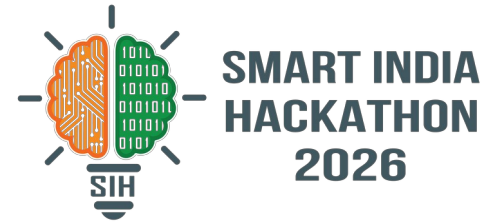
- Core pipeline (ingestion → ML → dashboard → alerts → reporting → routing) built on 16 NER villages; remaining work: swap simulated data for live feeds
- Govt API approval delays; sparse historical landslide records; connectivity gaps; low-literacy alert adoption
- Strategies: early API onboarding; offline fallback works with zero connectivity; voice alerts for low-literacy users

IMPACT AND BENEFITS



- Earlier warnings for admins, NDRF/SDRF and remote communities; reduces village isolation from blocked roads; real-time reporting tool for field officials
- Social: fewer casualties; Economic: less infrastructure damage, lower relief costs; Governance: data-driven, climate-resilient disaster planning

RESEARCH AND REFERENCES



- IMD (mausam.imd.gov.in) rainfall data; Bhuvan/ISRO geoportal for satellite & DEM data; GSI National Landslide Susceptibility Mapping & landslide inventory; NASA SMAP soil moisture; SIH 2026 PS 26001 problem statement (MDoNER)